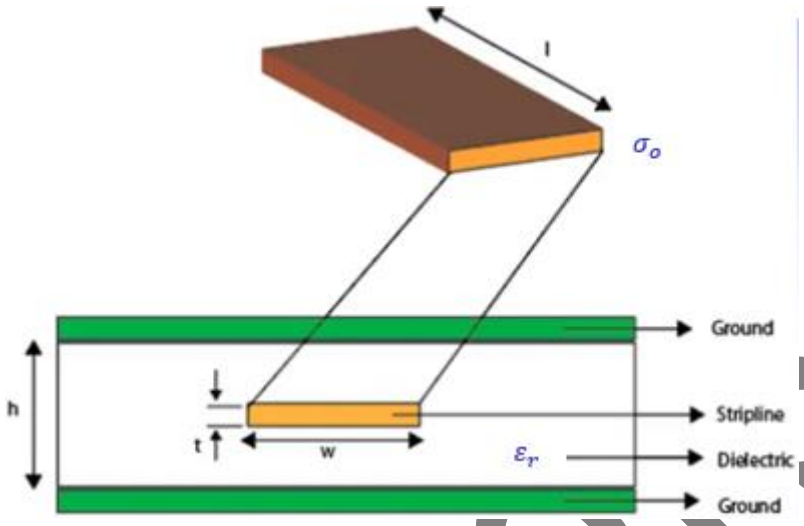
Model answer to Q 1(a): Discussion in Class

Sketch a symmetrical stripline, showing all relevant variables. Define each variable and give their units.



$$l, W, t, \sigma_o = length, width, thickness\ [m]\ and\ DC\ conductivity\ \left[\frac{S}{m}\right]\ of\ signal\ track$$

$$h = total\ thickness\ [m]\ and\ \varepsilon_r = effective\ relative\ permittivity\ of\ the\ dielectric$$

[5]

Model answer to Q 1(b): Discussion in Class

Compare and contrast stripline with microstrip, in terms of electromagnetic characteristics and practical implementations.

When compared to microstrip:

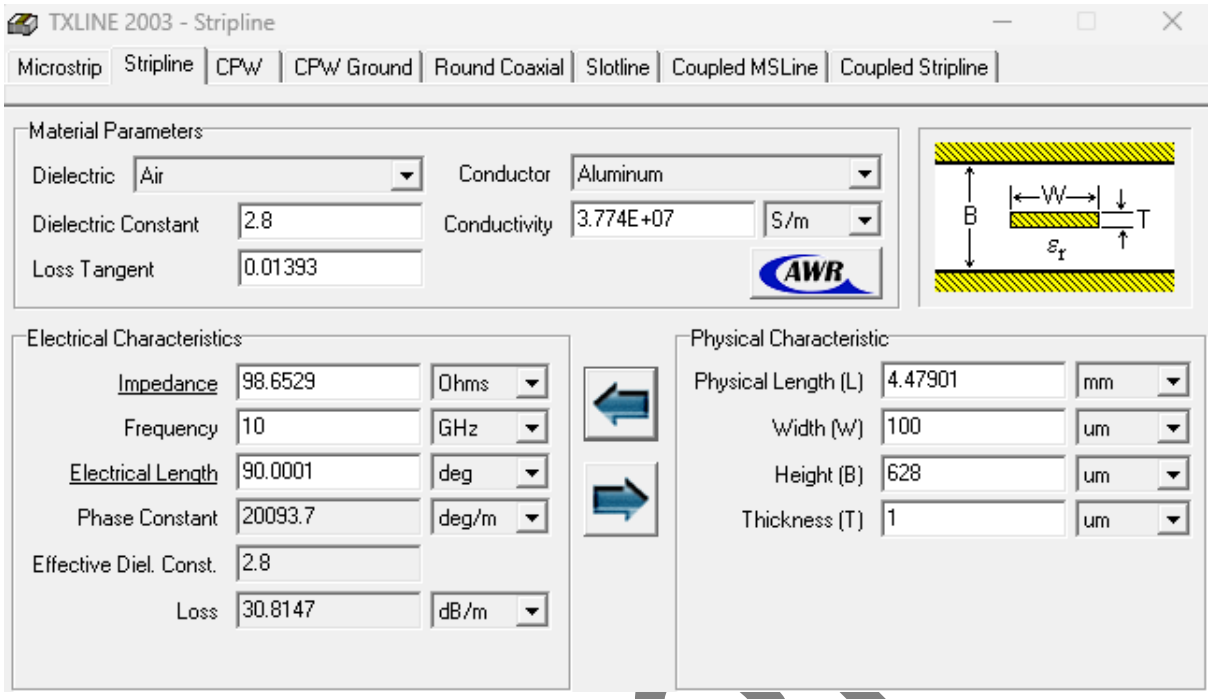
- no frequency dispersion with TEM-mode propagation and, thus, wider bandwidth
- narrower signal track width and, thus, higher sensitivity to physical variations
- more ohmic losses and, thus, lower unloaded Q-factor
- more fabrication processing and difficulty integrating lumped-element components

[5]

Model answer to Q 1(c): Bookwork and New Calculations

Assume any ground plane to be a perfect electrical conductor. The signal track has a width of 100 μm (equal to the total dielectric thickness divided by 2π) and its $skin\ depth \ll metal\ thickness \ll total\ dielectric\ thickness$. The signal track is aluminium (having a DC conductivity of $\sigma_o = 3.774 \times 10^7$ S/m) and the dielectric is polyimide (having an effective relative permittivity of $\varepsilon_r = 2.8 - j\ 0.039$). Unless otherwise stated, calculate the following at 10 GHz and include all units; clearly stating any simplifying assumptions:

Answers



i) Total dielectric thickness and effective signal track width.

$$W/h = 1/2\pi = 0.159155 \text{ and so } h = 628\ \mu\text{m}$$

since $t \ll h$: $$W_{eff}/h \approx W/h - \left(0.35 - W/h\right)^2 = 0.1227$$

$$W_{eff} = 0.1227\, h = 77.1\ \mu\text{m}$$

[3]

ii) Lossless characteristic impedance.

$$Z_o \approx \frac{\eta_o}{4\sqrt{\varepsilon_r}} \frac{1}{\left(W_{eff}/h + 0.441\right)} = 0.4435 \frac{\eta_o}{\sqrt{\varepsilon_r}} \approx 99.9\ \Omega$$

[2]

iii) Lossless phase velocity.

$$v_p \xrightarrow{Lossless} \frac{1}{\sqrt{LC}} = \frac{c}{\sqrt{\varepsilon_r}} \approx 179.28 \times 10^6\ \text{m/s}$$

[2]

iv) Lossless phase constant.

$$\beta = \omega\sqrt{LC} = \frac{\omega\sqrt{\varepsilon_r}}{c} = 350.47 \frac{\text{rad}}{\text{m}} = 20{,}080\ \text{deg/m}$$

[2]

v) Distributed capacitance.

$$\beta = \omega\sqrt{LC} = \omega C Z_0 \text{ and so } C = \frac{\beta}{\omega Z_0} = \frac{1}{v_p Z_0} = 55.83\ \text{pF/m}$$

[3]

vi) Distributed conductance.

Answer

$$tan\delta = \frac{0.039}{2.8} = 0.01393$$

$$G = tan\delta\ \omega C = 48.865\ \text{mS/m}$$

[3]

vii) Distributed inductance.

$$\beta = \omega\sqrt{LC} = \frac{\omega L}{Z_0} \text{ and so } L = \frac{\beta Z_0}{\omega} = \frac{Z_0}{v_p} = 557.23\ \text{nH/m}$$

[2]

viii) Skin depth in the signal track.

$$\delta s = \sqrt{\frac{2}{\omega\mu_o\sigma_o}} = 0.819\ \mu\text{m}$$

[2]

ix) Distributed resistance.

$$R \sim \frac{R_S}{2W} = \frac{1}{2\sigma_o\delta_o W} = 161.76\ \Omega/\text{m}$$

[2]

x) Lossy characteristic impedance.

$$Z_0 = \frac{R+j\omega L}{\gamma} = \sqrt{\frac{R+j\omega L}{G+j\omega C}} = 99.8987 + j0.46498\ \Omega$$

[3]

xi) Lossy propagation constant.

$$\gamma = \frac{R + j\omega L}{Z_0} = 3.25045\,\frac{\text{Np}}{\text{m}} + j\ 350.458\ \text{rad/m}$$

[3]

xii) Attenuation constant due to the conductors only.

$$\alpha_{conductor} = \frac{R}{2Z_0} = 0.8096\ \text{Np/m}$$

[2]

xiii) Attenuation constant due to the dielectric only.

$$\alpha_{dielectric} = \frac{Z_0 G}{2} = 2.4408\ \text{Np/m}$$

[2]

xiv) Total attenuation constant and insertion loss in dB/m.

$$\alpha = \alpha_{conductor} + \alpha_{dielectric} = \frac{R}{2Z_0} + \frac{Z_0 G}{2} = 3.25\ \text{Np/m}$$

$$Insertion\ loss = 8.686\ \alpha = 28.3\ \text{dB/m}$$

[3]

xv) Q-factor associated with the conductors only.

Answers

$$Q_{conductor} = \frac{\omega L}{R} = 216.44$$

[2]

xvi) Q-factor associated with the dielectric only.

$$Q_{dielectric} = \frac{\omega C}{G} = 71.78$$

[2]

xvii) Overall Q-factor.

$$Q = \frac{Q_{conductor} Q_{dielectric}}{Q_{conductor} + Q_{dielectric}} = 53.9$$

[2]

Answers

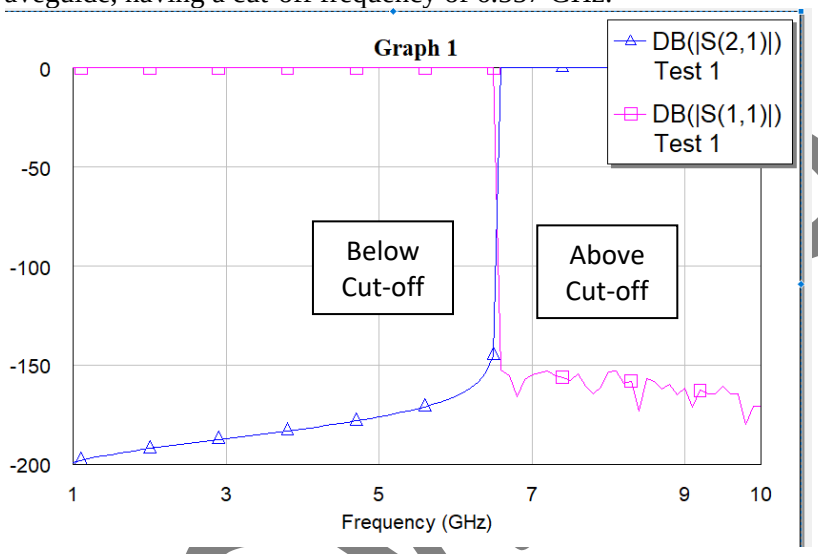
Model answer to Q 2: Bookwork and New Calculations

Consider two air-filled metal-pipe rectangular waveguide (MPRWG) standards, having the broad wall dimension *a* and narrow wall dimension *b*:

WR10: a = 2.540 mm and b = 1.270 mm
WR8: a = 2.032 mm and b = 1.016 mm

i) Sketch the typical S-parameter frequency characteristics for a typical low-loss MPRWG, clearly indicating the regions below and above its cut-off frequency.

For a WR90 waveguide, having a cut-off frequency of 6.557 GHz:



[4]

ii) Calculate the lowest cut-off frequencies for both the WR10 and WR8 MPRWGs.

$$\lambda_c = 2a \quad \therefore f_c = \frac{v_p}{\lambda_c} \Rightarrow \frac{c}{2a} \quad \text{for an air - filled waveguide}$$

fc = 59.055 GHz for RW10 and 73.819 GHz for WR8

[4]

iii) Calculate the guided-wavelength for the WR8 waveguide at 100 GHz.

$$\lambda_g = \frac{\lambda_o}{\sqrt{\mu'_r \varepsilon'_{reff} - \left(\frac{f_{cm0}}{f}\right)^2}}$$

= 4.447 mm

[4]

iv) If a WR8 MPRWG is connected between two WR10 MPRWGs, how long would the RW8 MPRWG need to be to excite the dominant Fabry-Perot resonance at 100 GHz?

Answers

$$length = \frac{\lambda_g}{2} = 2.224 \text{ mm}$$

[2]

v) For the Fabry-Perot resonator, described in 2(iv), what is the coupling mechanism?

Physically transitioning from a width of a = 2.540 mm to narrower a = 2.032 mm represents inductive coupling, while transitioning a height of b = 1.270 mm to a lower b = 1.016 mm represents capacitive coupling. As a result, there is combined inductive-capacitive electromagnetic coupling into and put of the resonator.

[3]

vi) Calculate the wave impedances for both the WR10 and WR8 MPRWGs at 100 GHz.

$$Z_{TEm0}^{air} = \frac{\eta_0}{\sqrt{1 - \left(\frac{f_{cm0}}{f}\right)^2}}$$

~ 467.2 Ω for RW10 and 558.8 Ω for WR8

[4]

vii) If a WR8 MPRWG is connected to a reference WR10 MPRWG feed line, at 100 GHz, calculate the voltage-wave reflection coefficient and return loss at the connecting flanges.

Voltage-wave Reflection Coefficient, $S_{11} = \frac{Z_{WR8} - Z_{WR10}}{Z_{WR8} + Z_{WR10}}$

= 0.0893

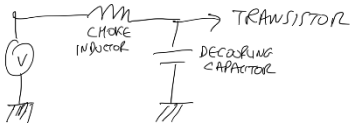
Return Loss, $RL_{IN} = -10 \log_{10}|S_{11}|^2$ = 21 dB

[4]

Answers

Model answer to Q 3: Bookwork and New Calculations

Consider the DC biasing of a microwave transistor. An ideal voltage-source power supply is connected to a shunt metal-insulator-metal decoupling capacitor, via a series choke inductor. Both the choke inductor and the decoupling capacitor are made from PEC. An ideal connection is made between the decoupling capacitor and the microwave transistor. Clearly state any assumptions made.



a) The choke inductor is made using a straight piece of 1 cm long and 1 mm diameter wire, suspended in air, having an inductance L_o. This wire is then encased within a zinc–nickel ferrite, having relative permeability $\mu_r = 3.1 - j\ 1.9$:

i) Assuming this permeability is frequency independent, sketch the equivalent circuit model for this choke inductor.
ii) Calculate all corresponding values at 1 GHz.
iii) Calculate the associated unloaded Q-factor Q_m from this circuit and compare this with that from the ferrite material.

a) $L_o = L_{ex} + L_{in}$

$L_o \simeq \frac{\mu_0}{2\pi} \ell \left[\ln\left\{\frac{4\ell}{d}\right\} - 1 + c \right]$ (L_{ex})

$L_{in} = 0$ since PEC

$\ell = 1\,cm \quad d = 1mm$

$\therefore L_o \simeq 5.38\,nH$

$Z_{SERIES} = j\omega L_o(\mu_r' - j\mu_r'') \equiv R + j\omega L$

$\mu_r' = 3.1 \quad \mu_r'' = 1.9$

$R(1\,GHz) = 64\,\Omega$ AND $L = 16.68\,nH$

$Q_m(1GHz) = \frac{\omega L}{R} = 1.64 \equiv \frac{\mu_r'}{\mu_r''} = 1.63$ ✓

[6]

b) The decoupling capacitor is formed using a 1 cm × 1 cm top plate and the bottom plate employs the ground plane on the other side of a 635 μm thick alumina substrate, having an effective relative permittivity of $\varepsilon_{reff} = 9.8 - j\ 0.0001$:

i) Assuming this permeability is frequency independent, sketch the equivalent circuit model for this decoupling capacitor.
ii) Calculate all corresponding values at 1 GHz.
iii) Calculate the associated unloaded Q-factor Q_m from this circuit and compare this with that from the substrate material.

Answers

$C_0 = \varepsilon_0 \frac{\text{AREA}}{\text{SEPARATION}}$

$\varepsilon_0 = 8.854\ pF/m$

$\text{AREA} = 1cm \times 1cm = 1\times10^{-4} m^2$

$\text{SEPARATION} = 635\times10^{-6}\ m$

$\therefore C_0 = 1.39\ pF$

$Y = j\omega C_0 (\varepsilon_r' - j\varepsilon_r'') = G + j\omega C$

$\varepsilon_r' = 9.8 \quad \varepsilon_r'' = 0.0001$

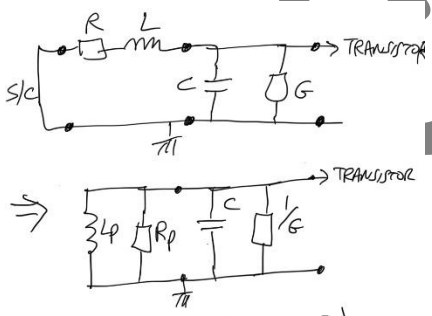
$G(1GHz) = 873.4\ nS$

AND $C = 13.62\ pF$

$Q_e = \frac{\omega C}{G} = 97,986 = \frac{\varepsilon'}{\varepsilon''} = 98,000$

[6]

c) By combining the equivalent circuit models for the power supply, choke inductor and decoupling capacitor, sketch the equivalent circuit model of the biasing network as an equivalent shunt *RLC*-network and calculate the driven resonance frequency. *Hint: first transform the series choke inductor model into its parallel equivalent circuit model and recalculate the associated resistances at the driven resonance frequency.*



$R_p = R(1 + Q_m^2)$

$\therefore R_p(1GHz) = 234\ \Omega$

$L_p = L\left(1 + \frac{1}{Q_m^2}\right)$

$\therefore L_p = 22.96\ nH$

DRIVEN RESONANCE FREQUENCY

$\omega_0 = \frac{1}{\sqrt{L_p C}} = 2\pi(284.6\ MHz)$

$R_p(284.6MHz) = \omega_0 L_0 \mu_r'' = 18.28(1+Q_m^2)$

$G(284.6MHz) = \omega_0 C_0 \varepsilon_r'' = 66.8\ \Omega$

$\therefore 1/G = 4\ M\Omega \quad \therefore R_p$ DOMINATES!

[6]

d) Using the *RLC* element values determined in 3(b) and 3(c):

i) Calculate the overall unloaded Q-factor Q_o of the network and compare this to that using the values of Q_m and Q_e (determined previously).

ii) Calculate the complex resonance frequency and from this calculate the unloaded Q-factor.

$$\tilde{\omega}_o = \omega_o \sqrt{1 - \frac{1}{\left(2Q_u(\omega_o)\right)^2}} + j\frac{\omega_o}{2Q_u(\omega_o)} \equiv \omega_o' + j\omega_o''$$

Answers

$$Q_U(\omega_0) = \omega_0 C R_p$$

$$R_p = 66.8\,\Omega \quad C = 13.62\,pF$$

$$\therefore Q_U(284.6\,MHz) = 1.63$$

$$\text{BUT } Q_U = \frac{Q_m Q_e}{Q_m + Q_e} \cong Q_m = 1.63 \checkmark$$

$$\tilde{f}_0 = f_0(0.952 + j\,0.307) = f_0' + jf_0''$$

$$\therefore \tilde{f}_0 = 271\,MHz + j\,87.4\,MHz$$

$$Q_U(f_0) = \frac{f_0}{2f_0''} = 1.63 \checkmark$$

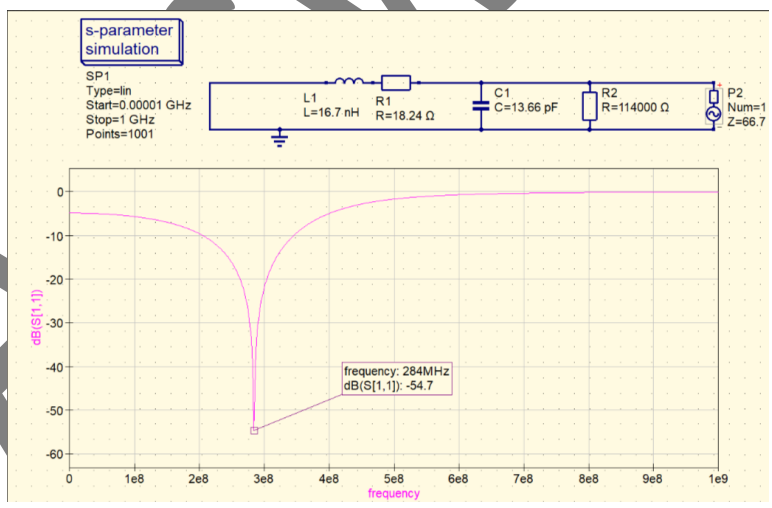
[4]

e) Using the complex resonance frequency determined in 3(c), at the driven resonance frequency, calculate the return loss for the DC biasing network 'seen' by the microwave transistor having an impedance of 66.7 Ω. What can be inferred, in terms of the useful bandwidth of the microwave circuit?

$$\rho = \frac{Z - Z_0}{Z + Z_0} = \frac{66.8 - 66.7}{66.8 + 66.7} @ f_0$$

$$\therefore \rho(f_0) = 7.49 \times 10^{-4}$$

$$\text{RETURN LOSS} = -20\log_{10}\{\rho(f_0)\} = 62.5\,dB$$



For any RF signal at the transistor that 'sees' the biasing network an almost perfect open circuit if found above approximately 800 MHz; while a perfect absorber is found at f_o, with maximum RF signal power absorbing into the biasing network when the transistor impedance is equal to R_p

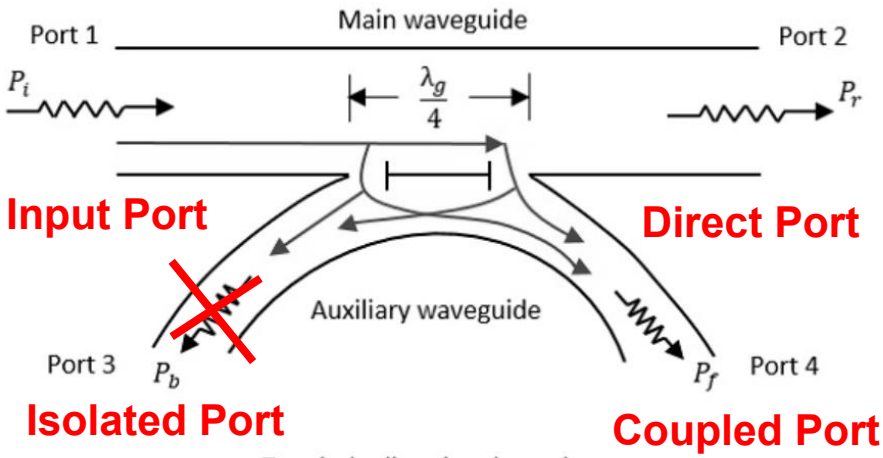
[3]

Model answer to Q 4: Bookwork interpretation

This question relates to directional couplers.

a) For a 3 dB waveguide coupler:

i) Sketch the coupler in rectangular waveguide, clearly indicating the port numbers and give the usual name attributed to each port. Briefly describe how it works.



Two-hole directional coupler

Holes spaced λg/4 apar nd ideally cancel out-of-phase in the opposite direction towards the isolated Port 3

[3]

ii) By inspection, or otherwise, write down the associated lossless S-parameter matrices for the centre frequency of operation.

$$[S]_{waveguide} = \frac{-j}{\sqrt{2}}\begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix} @\ \omega_0$$

[1]

iii) What is the requirement for the port termination impedances?

All ports must be terminated at the wave impedance for the waveguide, which is frequency dependent.

[2]

iv) State at least two applications for this coupler.

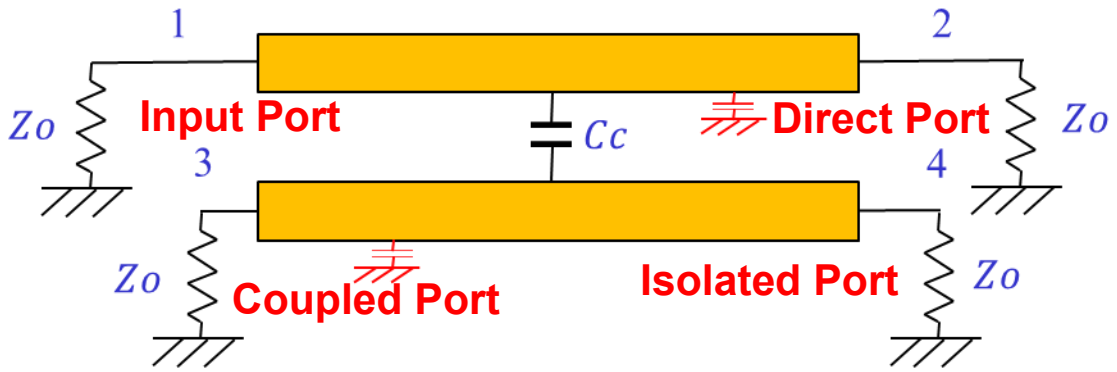
This is a very low loss coupler that can be used for very low signal applications (e.g. radio astronomy) or very high power applications (e.g. radar).

[2]

b) For a 3 dB coupled-line coupler:

i) Sketch the coupler in microstrip, clearly indicating the port numbers and give the usual name attributed to each port.

Answer



[2]

ii) By inspection, or otherwise, write down the associated lossless S-parameter matrices for the centre frequency of operation.

$$[S]_{coupled-line} = \frac{-j}{\sqrt{2}}\begin{pmatrix} 0 & 1 & j & 0 \\ 1 & 0 & 0 & j \\ j & 0 & 0 & 1 \\ 0 & j & 1 & 0 \end{pmatrix} @\ \omega_0$$

[1]

iii) What is the requirement for the port termination impedances?

All ports must be terminated at the reference impedance, which is frequency independent.

[2]

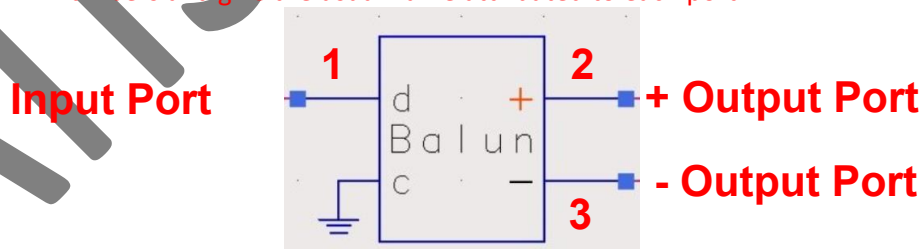
iv) State at least two applications for the this coupler.

This is a relatively lossy loss coupler that can be used for medium signal power level applications that require two signals at the same power levels but 90 degrees apart (e.g. Reflection topology circuits and balanced topology circuits).

[2]

c) For a 3 dB Marchand Balun:

v) Sketch its symbolic representation, clearly indicating the port numbers and give the usual name attributed to each port.



[1]

i) By inspection, or otherwise, write down the associated lossless S-parameter matrices for the centre frequency of operation.

$$[S]_{Marchand} = \frac{-j}{\sqrt{2}}\begin{pmatrix} 0 & -1 & 1 \\ -1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} @\ \omega_0$$

[1]

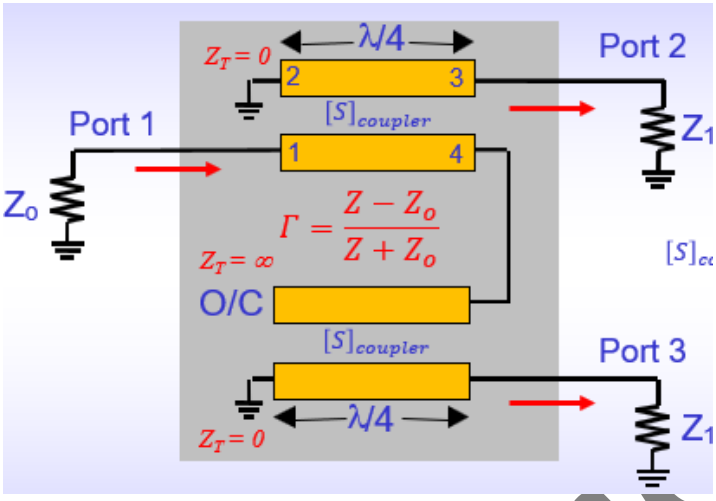
ii) What is the requirement for the port termination impedances?

This lossless 3-port network cannot be matched at all ports and two ports are not isolated from one another

[1]

iii) Using coupled-line couplers, show how the balun can be implemented by sketching its complete circuit.

Answers



[3]

iv) How can the coupler in 4(b) be physically modified in order to be used in the balun?

Both coupled-line coupler need to have 4.44 dB of coupling. This means that the lines must be separated in order to reduce the level of coupling from 3 dB. However, since the line impedance will increase then the lines also need to be made wider to achieve the impedance matching.

[2]

v) State at least two applications for this balun.

Baluns are use create two anti-phase signals of equal power (e.g. mixers and dipole antennas).

[2]

Answers